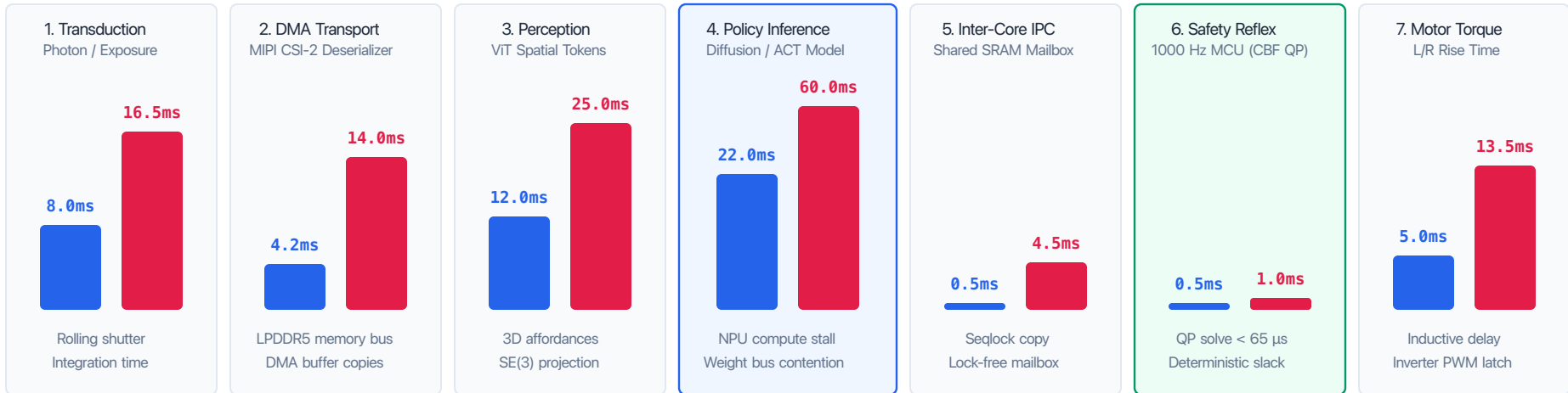


THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Deconstructing where time is lost: Nominal P50 (52.2 ms) vs. P99 Latency Tail Blowout (126.0 ms) Breaching World Deadlines



■ P50 Nominal Execution

■ P99 Contention & Jitter Tail Blowout

----- Physical World Stability Deadline ($\tau_{\text{world}} = 100 \text{ ms}$)

P50 Total: 52.2 ms (Safe Closed-Loop Margin: +47.8 ms)

P99 Total: 126.0 ms (BREACH!)

$\tau_{\text{world}} = 100 \text{ ms}$

Systems Insight:

While mean latency ($P50 = 52.2 \text{ ms}$) looks safe in benchmarking, real-world memory bus contention and OS scheduling inflate $P99$ past the world deadline, resulting in open-loop drift and physical